GP's predictions.

4.2.2 Computing the probability of improvement

Having isolated what we aim to achieve in BayesOpt—namely, improving from the current incumbent—we are now ready to finally begin discussing the BayesOpt policies that seek to achieve this goal. In this subsection, we will learn about *Probability of Improvement*, which is a policy that measures how likely a candidate point is able to improve.

The idea of measuring how likely it is that a candidate point improves from the incumbent corresponds to the probability a point is a "good" one according to Figure 4.7. This was also hinted in Section 6.2.1 with the GP, where we said that:

1. The point at 0 (Figure 4.9) has a large portion of its normal distribution highlighted as possible improvements. In other words, it has a high probability of improving.
2. The point at -2 (center panel of Figure 4.10) has a 0.5 probability of improving, as only half of its normal distribution exceeds the incumbent.
3. The point at 3 (right panel of Figure 4.10) has most of its normal distribution below the threshold, so it has a very low probability of improving.

We can make this computation more concrete by noticing that the probability that a point improves from the incumbent is equal to the area of the highlighted portion in Figure 4.9 and Figure 4.10.

IMPORTANT **Area under the probability density curve**

The entire area under any normal curve is 1, so the area of the highlighted portion in Figure 4.9 and Figure 4.10 exactly measures how likely it is that the normal random variable in question (the function value at a given point) exceeds the incumbent.

Thanks to how mathematically convenient normal distributions and GPs are, we can easily compute the area of this highlighted region using the cumulative density function (CDF). Computationally, we can make use of PyTorch's `torch.distributions.Normal` class, which implements normal distributions and offers the useful `cdf()` method. Specifically, suppose we are interested in calculating how likely the point at 0 is able to improve from the incumbent. We will follow the procedure describe in Figure 4.11:

1. First, we use the GP to compute the mean and standard deviation predictions at the point at 0.
2. We then compute the area under the normal curve defined by the mean and standard deviation above with the incumbent being the cutoff. We use the CDF for this computation.
3. Finally, we subtract the CDF value from 1 to obtain the probability of improvement for the candidate point.

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